Bangalore Rapido Ride Analysis

Understanding Demand, Cancellations & Revenue

Sector: Transportation / Ride-Hailing

Team ID: G-1

Team Members & Roles:

- **Aniket Pathak:** Project Lead (Statistical Analysis & Tableau Dashboard)
- **Nipun Patlori:** Data Lead (ETL & Cleaning)
- **Saswataduity Bhuin:** ETL Lead (Dashboard Support)
- **Akshay Y:** Analysis Lead
- **Narendra Singh:** Visualization Lead
- **Sayan Bhattacharya:** Strategy Lead
- **Vedant Madne:** PPT and Quality Lead

Faculty Mentor: Archit Raj, Ayushi Vashishth

Institute: Newton School of Technology, Rishihood University

GitHub Repository: aniket-bit7/SectionA_G1_Rapido_Ride_Analysis

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Executive Summary

Problem: Rapido faces a significant operational challenge in Bangalore: balancing ride demand with driver supply. High cancellation rates—currently at **10.4%**—erode customer trust and lead to substantial revenue leakage during peak hours.

Approach: Our team utilized a Python-based ETL pipeline to clean and transform 50,000 ride transactions. We engineered features such as `demand_intensity` and `peak_hour` flags to perform statistical correlation analysis, followed by the development of an interactive Tableau dashboard to visualize geographic and temporal hotspots.

Key Insights:

- **Supply Bottlenecks:** Areas with high demand (z-score > 1.0) experience significantly higher cancellation rates, confirming that supply cannot keep up with peak volume.
- **Peak Vulnerability:** Cancellations spike by **15-20%** during morning (8-10 AM) and evening (5-7 PM) windows.
- **Service Resilience:** Two-wheeler services (`bike`) maintain the highest completion rates (>92%) even during heavy traffic.

Key Recommendations:

- Implement targeted driver incentives in high-demand zones like **Koramangala** and **HSR Layout** during peak windows.
- Optimizing supply in the top 5 "High Demand" zones could reclaim an estimated **₹120,000** in monthly revenue.

Sector Context & Problem Statement

Sector & Business Context: The ride-hailing industry in Bangalore is highly competitive and plagued by extreme traffic congestion. This project serves the **Operations Manager**, who needs to maximize revenue by reducing the number of unfulfilled ride requests. Solving this problem provides direct business value by increasing fleet efficiency and customer retention.

Problem Statement: How do service type, time of day, and location influence ride cancellation rates and revenue?

Project Objectives:

- Quantify the relationship between demand intensity and cancellation rates.
- Identify high-revenue zones suffering from supply shortages.
- Develop a KPI framework to monitor operational health.

Success Criteria: The project is successful if we identify at least 3 actionable geographic bottlenecks where driver reallocation can improve completion rates.

Data Description & Cleaning

Data Description

- **Dataset Overview**: The analysis uses a transactional dataset of 49,801 cleaned ride records from Bangalore.
- **Temporal Scope**: Data spans from **April 10, 2024, to April 29, 2024**, covering peak and off-peak demand cycles.
- **Key Attributes**:
  - **Dimensions**: `ride_status` (Completed/Cancelled), `services` (Bike, Auto, Cab), and geographic markers (`source`, `destination`).
  - **Metrics**: `distance`, `duration`, `ride_charge`, and `total_fare`.
- **Limitations**: The dataset is simulated and lacks specific qualitative reasons for cancellations (e.g., traffic vs. driver behavior).

Data Cleaning & ETL Pipeline

The ETL process focused on ensuring financial accuracy and removing physical impossibilities from the dataset:

- **Financial Handling**: For cancelled rides, `total_fare` and `ride_charge` were set to **0** to prevent skewing revenue averages while retaining the record for cancellation analysis.
- **Data Validation**:
  - **Distance**: Removed rides with **0 km** or **>100 km** to eliminate GPS anomalies.
  - **Duration**: Dropped records with durations **≤ 0 minutes**.
  - **Deduplication**: Cleaned redundant entries based on unique `ride_id`.
- **Standardization**: Lowercased and stripped whitespace from all categorical text (e.g., mapping `bike lite` to `bike_lite`) to ensure consistent grouping.
- **Feature Engineering**:
  - **Temporal**: Extracted `hour`, `day_of_week`, and created a binary `peak_hour` flag (8-11 AM, 5-8 PM).
  - **Efficiency Metrics**: Calculated `avg_speed_kmh` and `fare_per_km` to evaluate service performance and unit economics.

KPI & Metric Framework

To evaluate the operational efficiency and financial health of Rapido's Bangalore market, we established a multi-layered KPI framework. These metrics transition from high-level business health to granular unit economics.

7.1 Primary Business KPIs

- **Ride Cancellation Rate (10.4%)**: Calculated as the ratio of cancelled rides to total bookings (5,180 / 49,801). This serves as the primary proxy for "Lost Opportunity." A double-digit rate confirms a significant supply-demand mismatch during peak intervals.
- **Total Gross Revenue (₹6,837,929)**: The aggregate of `total_fare` from all successfully completed trips during the 20-day analysis window.
- **Average Revenue per Ride (ARPR - ₹153.24)**: This metric evaluates the average ticket size per completion. It is used to benchmark the performance of premium services (Cabs) against high-volume services (Bikes).

7.2 Operational Efficiency Metrics

Beyond top-line revenue, we utilized derived metrics to understand service quality:

- **Demand Intensity (Z-Score)**: A normalized metric created to identify "High Demand" clusters. By mapping this against cancellation rates, we statistically validated that hotspots like **Koramangala** suffer the most from driver shortages.
- **Avg Speed (km/h)**: Derived from `distance / duration`. This allows the Operations Team to see how Bangalore's traffic congestion directly correlates with increased cancellation rates for four-wheelers compared to bikes.
- **Fare per KM (Unit Economics)**: Measures the revenue density of a trip. This metric helps identify if certain routes are being underpriced relative to the time and effort required by drivers.

7.3 Strategic Mapping

- **Objective Alignment**: The **Cancellation Rate** is mapped directly to our "Success Criteria". A reduction in this KPI in high-demand zones is the primary goal of our recommendations.
- **Decision Support**: These KPIs provide the **Operations Manager** with a quantitative basis for dynamic pricing and localized driver incentives.

Exploratory Data Analysis (EDA)

Service Distribution & Revenue Share

The Bangalore market shows a clear preference for two-wheeler mobility.

- Volume Leaders: Bike and Bike_lite combined account for 35.1% of all bookings. These services act as the backbone of high-frequency short-distance travel.
- Revenue Contributors: While Cab_economy and Auto have lower booking volumes, they contribute disproportionately to the total revenue due to higher base fares and longer average trip distances.

Correlation & Statistical Relationships

We examined the drivers of pricing and reliability through correlation matrices:

- Distance vs. Fare: A strong Pearson correlation of $r = +0.798$ was observed between distance and total_fare. This confirms that the pricing engine is functioning correctly based on trip length.
- Demand vs. Reliability: A distinct negative correlation exists between demand_intensity and the completed flag. As demand intensity (z-score) exceeds 1.0, the probability of cancellation increases by nearly 22%, confirming a supply-side bottleneck in the most profitable areas.

8.3 Temporal & Peak Hour Trends

Ride dynamics in Bangalore are highly sensitive to the city's "Rush Hour" cycles:

- Morning Surge: Ride volume peaks sharply between 8:00 AM and 10:59 AM. During this window, the cancellation rate for Cab_economy is 18% higher than the daily average, likely due to drivers avoiding heavy traffic corridors.
- Evening Dynamics: The evening peak (5:00 PM – 7:59 PM) shows a different pattern where Auto cancellations spike, suggesting drivers prefer shorter "last-mile" trips over long-distance commutes as the day ends.
- Weekend Shifts: Weekend demand is more evenly distributed throughout the day, with a significant increase in Parcel and Economy service demand during late afternoon hours (2 PM – 4 PM).

8.4 Geographic Hotspots

Analysis of source and destination clusters reveals high-intensity zones:

- Primary Hubs: Areas like Koramangala, HSR Layout, and Indiranagar generate approximately 40% of total revenue.
- Cancellation Hotspots: These same revenue hubs exhibit the highest source_cancel_rate, indicating that customers in the most lucrative zones are also the most likely to experience service failures.

8.5 Operational Efficiency (Avg Speed)

- Congestion Impact: The avg_speed_kmh for completed rides drops by an average of 35% during peak hours compared to late-night slots.
- Service Resilience: Bike services maintain a 12% higher average speed during peak traffic than four-wheelers, explaining their higher completion rates and reliability during the 8–10 AM window.

Statistical Analysis & Hypothesis Testing

To move beyond observation into validation, we applied inferential statistics to confirm the drivers of system inefficiency.

Hypothesis Testing (Chi-Squared Validation)

- **Objective**: To statistically confirm if temporal factors (peak vs. off-peak) influence service reliability.
- **Testing**: We conducted a **Pearson's Chi-squared test** on `ride_status` and the `peak_hour` flag.
- **Results**: The test yielded a **p-value < 0.05**, allowing us to reject the null hypothesis of independence.
- **Business Conclusion**: Cancellations are not random; they are statistically tied to peak traffic windows, confirming that Rapido's current supply model fails during high-congestion periods.

Segmentation via Demand Intensity

- **Methodology**: We utilized **z-score normalization** on ride volumes to segment Bangalore into "High Demand" ($z > 1.0$) and "Low Demand" ($z < 0$) zones.
- **Insight**: High Demand zones (e.g., Koramangala, HSR Layout) account for roughly **40% of total revenue** but suffer from disproportionately high `source_cancel_rate`.
- **Strategic Value**: This segmentation allows for targeted resource allocation rather than a "one-size-fits-all" city-wide strategy.

Root Cause Analysis: The Supply Bottleneck

- **The "Demand Paradox"**: Analysis confirms that zones with the highest customer interest (demand intensity) also have the lowest completion probability.
- **Revenue Leakage**: Statistical modeling suggests that this is a **driver-side bottleneck**. Drivers likely "cherry-pick" or avoid these zones during peaks due to high traffic and lower `avg_speed_kmh`.
- **Financial Impact**: By correlating these bottlenecks with `total_fare`, we estimated that a **10% reduction** in cancellations in these high-potential segments could reclaim **₹120,000 monthly**.

Tableau Dashboard Design

The dashboard was developed to bridge the gap between raw data and operational execution, providing the Rapido Operations Team with a diagnostic tool to monitor and mitigate revenue leakage.

10.1 Dashboard Objectives

- **Quantify Revenue Leakage**: Visualize the financial impact of cancellations across different city zones.
- **Supply-Demand Alignment**: Identify specific time windows where service availability fails to meet ride demand.
- **Resource Optimization**: Enable managers to reallocate the fleet based on real-time "hotspot" identification.

10.2 View Structure & Visual Hierarchy

- **Executive Summary View**:
  - **Headline KPIs**: Real-time display of **Total Revenue (₹6.84M)**, **Overall Cancellation Rate (10.4%)**, and **Total Completed Rides**.
  - **Revenue by Service Type**: A bar chart comparing the financial contribution of `Bike`, `Auto`, and `Cab_economy`, allowing for quick identification of the most profitable segments.
  - **Temporal Trend Line**: A dual-axis chart showing ride volume vs. cancellation rate over the 20-day period.
- **Operational Drill-Down View**:
  - **Geographic Heatmap**: A map-based visualization using `source` coordinates to identify "Hotspots" such as **Koramangala** and **HSR Layout**.
  - **Demand Intensity Matrix**: A scatter plot mapping `demand_intensity` against `source_cancel_rate` to highlight areas with critical supply shortages.
  - **Peak vs. Off-Peak Comparison**: Side-by-side analysis of performance during the 8–11 AM and 5–8 PM windows.

10.3 Interactive Elements & Navigation

To ensure the dashboard is actionable for different stakeholders, we implemented the following interactive features:

- **Global Filters**: Users can slice the entire dataset by **Service Type** (e.g., isolating `bike_lite` performance) or **Payment Method**.

- **Time Slider**: Allows for granular analysis of specific dates or peak/off-peak hour segments.
- **Dynamic Tooltips**: Hovering over geographic hotspots reveals specific metrics such as `avg_speed_kmh` and `fare_per_km` for that zone, providing immediate root-cause context for cancellations.

10.4 Technical Implementation

- **Data Integration**: The dashboard is powered by the `cleaned_rides.csv` output from the Python ETL pipeline, ensuring data consistency between statistical reports and visual outputs.
- **Accessibility**: The dashboard is hosted on **Tableau Public** for stakeholder access.

Insights Summary & Impact

The analysis of 49,801 ride transactions reveals critical systemic gaps in Bangalore's ride-hailing supply chain.

- **The "High-Demand Cancellation Trap"**: Statistical validation (z-score > 1.0) confirms that areas with the highest booking volumes experience significantly higher cancellation rates. This indicates that Rapido is failing precisely where the market is most lucrative.
- **Peak Hour Revenue Leakage**: Cancellations spike by **15–20%** during morning (8–10 AM) and evening (5–7 PM) windows. This peak-hour penalty results in approximately **₹4,000 in lost revenue daily**, primarily affecting `cab_economy` and `auto` services.
- **Service-Level Resilience**: `Bike` and `bike_lite` services maintain completion rates above **92%** even during peak congestion. These services are the most resilient assets in the fleet, suggesting they should be the primary focus during heavy traffic periods.
- **Geographic Revenue Concentration**: A small number of clusters—specifically **Koramangala and HSR Layout**—generate **40% of total revenue**. However, these zones suffer from inconsistent driver availability, making them high-priority targets for supply intervention.

Impact Estimation

By addressing these insights through targeted operational shifts, Rapido can realize substantial financial and efficiency gains.

- **Annual Revenue Recovery**: Reducing cancellations by just **10%** in the top 5 high-demand zones would reclaim approximately **₹120,000 per month**, scaling to **₹1.44M annually**.
- **Operational Efficiency Gains**: Implementing geographic driver pre-positioning—moving supply toward high-intensity source zones 30 minutes before peak windows—is estimated to reduce average customer wait times by **15%**.
- **Strategic Reliability**: Improving completion rates in revenue hotspots directly enhances customer lifetime value (CLV). By stabilizing the "cancellation trap" areas, Rapido can reduce customer churn to competing platforms during rush hours.

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Recommendations & Conclusion

To mitigate the identified "Cancellation Trap" and recover lost revenue, we propose a three-tiered intervention strategy focused on supply-side elasticity.

- **Tier 1: Targeted Geographic Incentives**:
  - **Action**: Implement a "Peak-Zone Completion Bonus" for drivers operating in **Koramangala, HSR Layout, and Indiranagar**.
  - **Rationale**: Since these zones generate 40% of revenue but suffer from the highest supply bottlenecks, localized incentives will discourage drivers from "cherry-picking" less congested peripheral areas during rush hours.
- **Tier 2: Dynamic Supply Rebalancing**:
  - **Action**: Utilize the "Bike" and "Bike_lite" fleet as the primary resilience tool during the 8–10 AM morning surge.
  - **Rationale**: Analysis confirms that two-wheelers maintain a **12% higher average speed** and **92%+ completion rate** during heavy traffic. Reallocating these drivers from peripheral "Low Demand" zones to central hubs will maximize throughput.
- **Tier 3: Service-Specific Surge Optimization**:
  - **Action**: Apply aggressive surge pricing specifically for `cab_economy` and `auto` during peak evening windows (5–8 PM).
  - **Rationale**: The statistically significant p-value (< 0.05) for peak-hour cancellations suggests that current fare structures do not sufficiently compensate drivers for the increased duration and effort required in traffic.

Impact Estimation & Value Proposition

- **Revenue Recovery**: By stabilizing supply in the top 5 revenue-generating zones, Rapido can reclaim an estimated **₹120,000 monthly** (₹1.44M annually) in previously lost fares.
- **Operational Reliability**: These interventions are projected to reduce average customer wait times by **15%** and lower the aggregate cancellation rate from **10.4%** toward a target of **8%**.

Conclusion

This project successfully decoded the correlation between demand intensity, geographic hotspots, and service reliability in the Bangalore market. The analysis proved that Rapido's primary challenge is not a lack of demand, but a **localized supply bottleneck** during peak temporal windows.

By transitioning from a uniform city-wide operation to a **data-driven, zone-specific allocation strategy**, the Operations Team can bridge the supply-demand gap. Implementing these recommendations will not only recover significant revenue leakage but also strengthen customer loyalty by providing a more reliable service in the city's most critical high-intensity hubs.

Appendix & Contribution Matrix

- **Data Dictionary:** Full definitions for `demand_intensity`, `peak_hour`, and `fare_per_km`.
- **Cleaning Log:** Documented in `notebooks/02_cleaning.ipynb`.

Contribution Matrix:

Team Member	Sourcing	ETL	Analysis	Stats	Tableau	Report
Aniket Pathak	Owner	Support	Support	Owner	Owner	Support
Nipun Patlori	Support	Owner	Owner	Support	Support	Support
Saswata B.	Support	Support	Support	Support	Owner	Support
Akshay Y	Support	Support	Support	Owner	Owner	Support
Narendra S.	Support	Support	Support	Support	Owner	Support
Sayan B.	Support	Support	Support	Support	Support	Support
Vedant M.	Support	Support	Support	Support	Support	Support

Declaration: We confirm that these details are accurate and verifiable via GitHub. **Team Lead Signature:** *Aniket Pathak* **Date:** April 29, 2026